We have performed the following stages to create the data visualisation

1.Dataset Download:

Download the Electric Vehicle Charge and Range Analysis dataset in CSV format from the provided project repository or a trusted open-data source. Store the dataset in a dedicated project folder for easy access and management.

2. Data Import:

import the dataset into Tableau or Microsoft Excel. Ensure all records and fields such as Vehicle Model, Battery Capacity (kWh), Charging Time, Charging Station Type, Range (km), Charging Speed, Vehicle Type, State, and Charging Cost are imported correctly.

3. Data Inspection:

Examine the dataset to understand its structure, including the number of rows, columns, data types, and available variables. Identify important attributes such as battery capacity, charging duration, driving range, charging station type, and vehicle category.

4. Data Cleaning:

Remove duplicate records, handle missing values, correct formatting inconsistencies, standardize categorical values (e.g., AC Charger, DC Fast Charger, Level 1, Level 2), and ensure numerical fields such as battery capacity, charging time, charging cost, and vehicle range are stored correctly.

5. Data Validation:

Validate the dataset by checking for incorrect or inconsistent values. Confirm that battery capacities, charging times, vehicle ranges, and charging costs fall within realistic and acceptable limits.

6. Data Transformation:

Create calculated fields such as Charging Efficiency, Estimated Charging Cost, Range per kWh, or Battery Performance Category. Rename columns for clarity, standardize text values, and prepare the dataset for meaningful analysis.

7. Data Filtering:

Filter the data based on relevant criteria such as Vehicle Type, Battery Capacity, Charging Station Type, State, Charging Speed, Range Category, or Charging Time to focus on specific EV charging and performance trends.

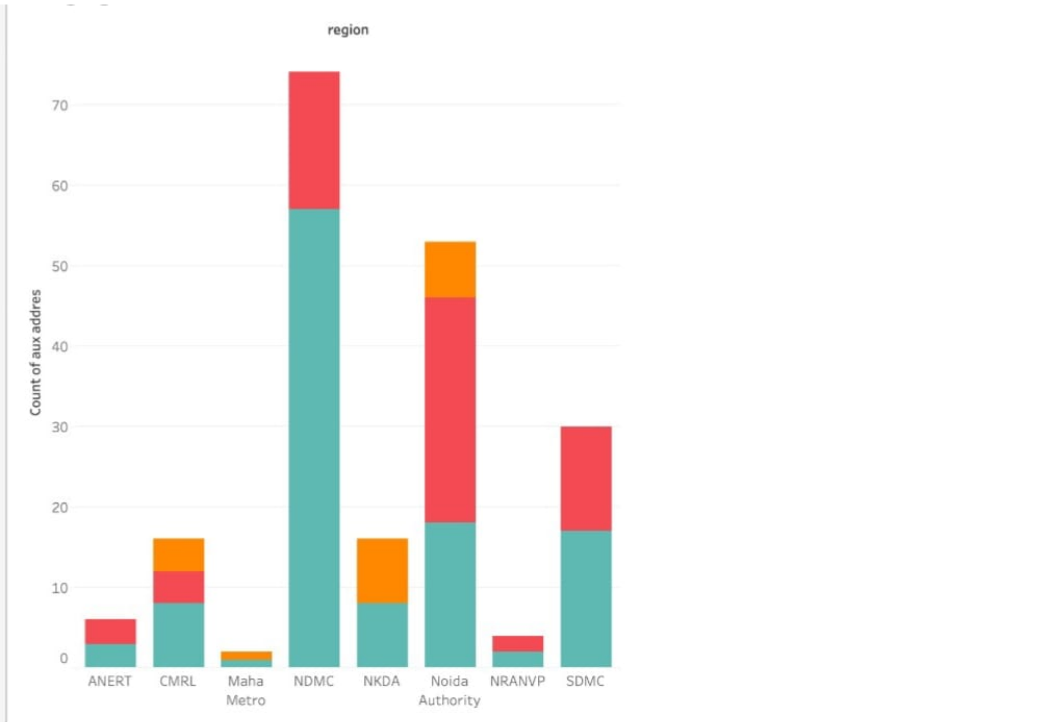
8. Data Verification:

Perform a final quality check to ensure the processed dataset is accurate, complete, consistent, and suitable for visualization. Verify that all transformations, calculations, and filters have been applied correctly.

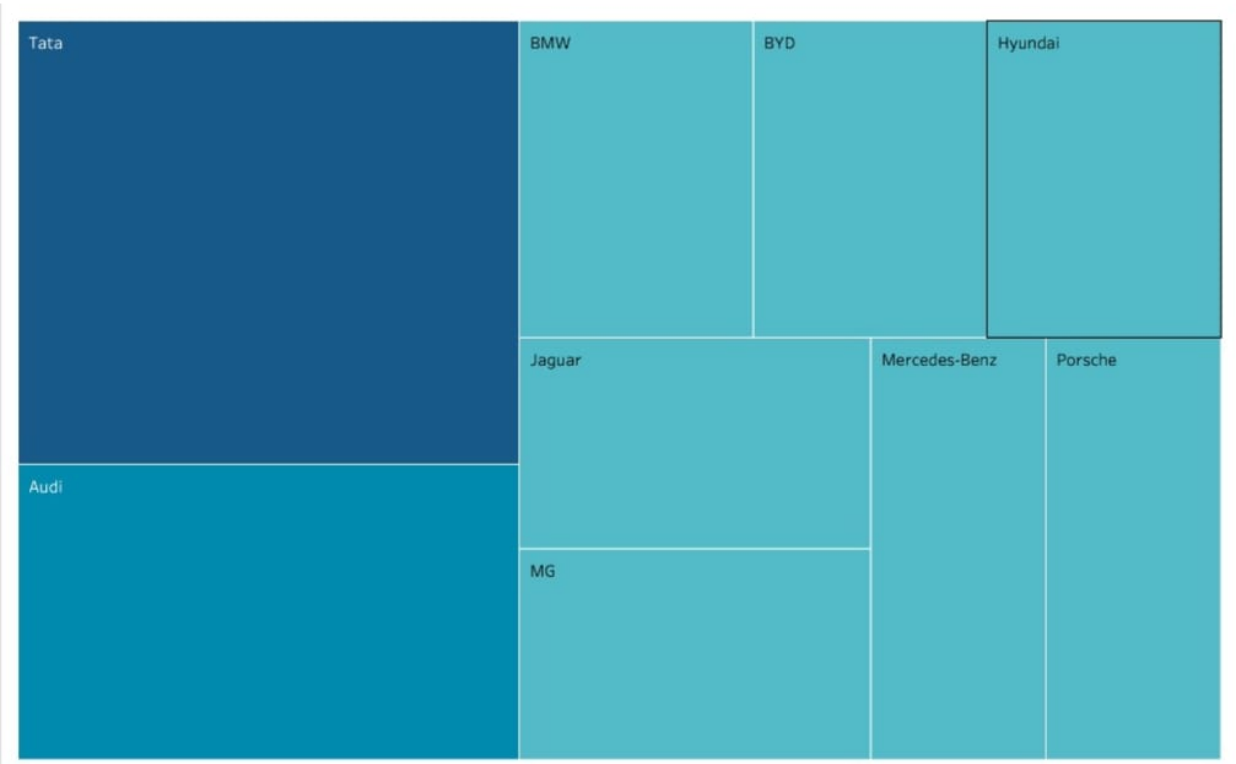
9. Data Visualization Preparation:

Load the cleaned dataset into Tableau, assign appropriate data types, create relationships if required, and develop interactive charts, dashboards, and visualizations to analyze electric vehicle charging.

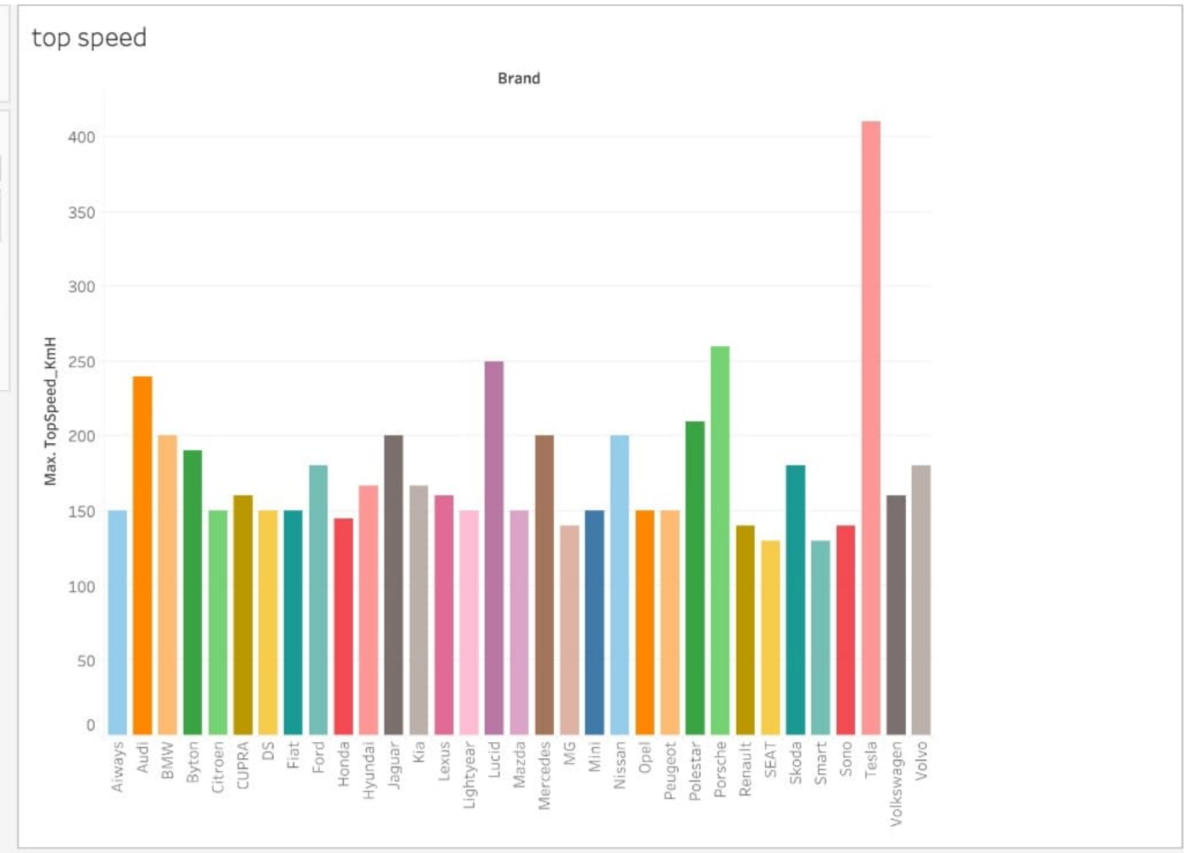
Business Question-1: What regional trends can be identified in EV charging station availability to support future expansion planning?



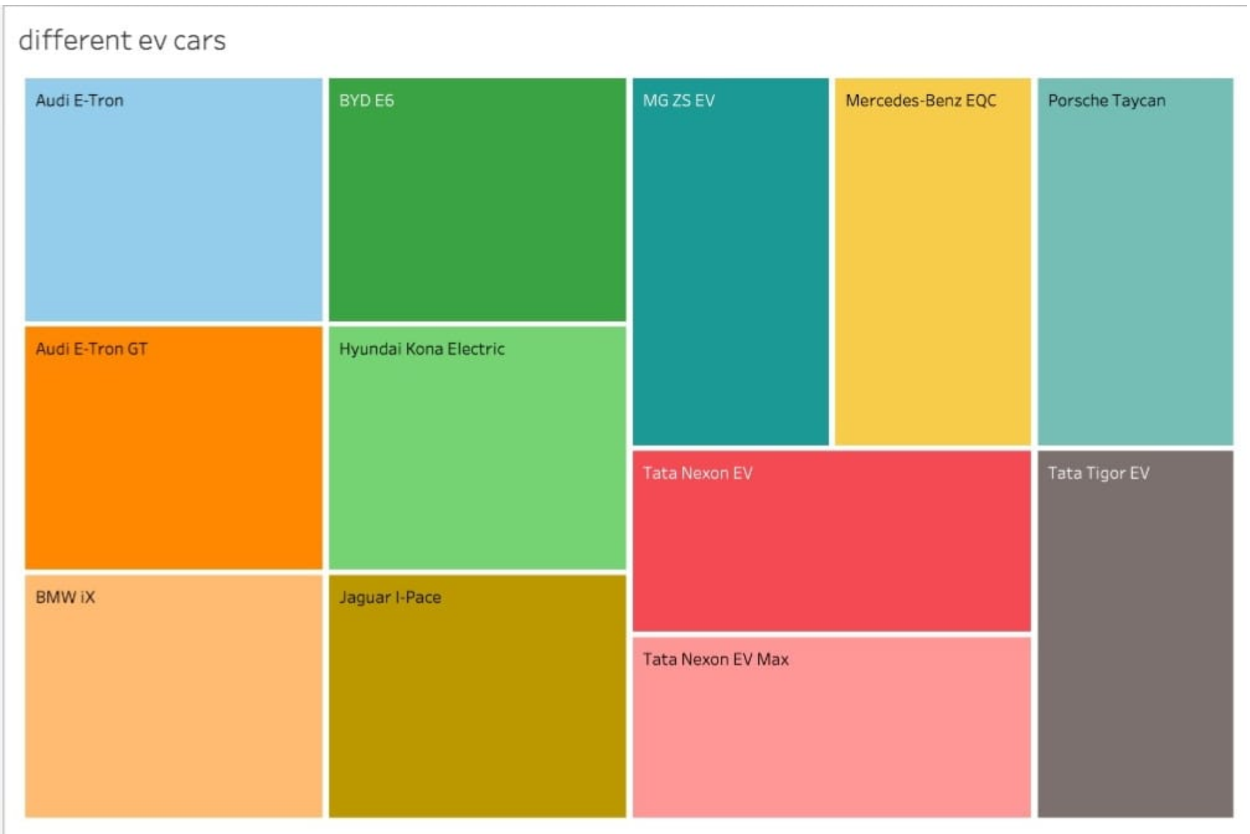
Business Question-2: Which electric vehicle brand has the highest market presence, and how is the distribution of vehicles across different EV manufacturers?



Business Question-3: Which electric vehicle brand offers the highest top speed, and how does maximum speed vary across different EV manufacturers?



Business Question-4: Which electric vehicle models have the highest representation in the dataset, and how is the distribution of different EV models across the market?



Business Question-5: Which body style is most commonly offered by electric vehicle brands, and how do different manufacturers distribute their EV models across body styles?